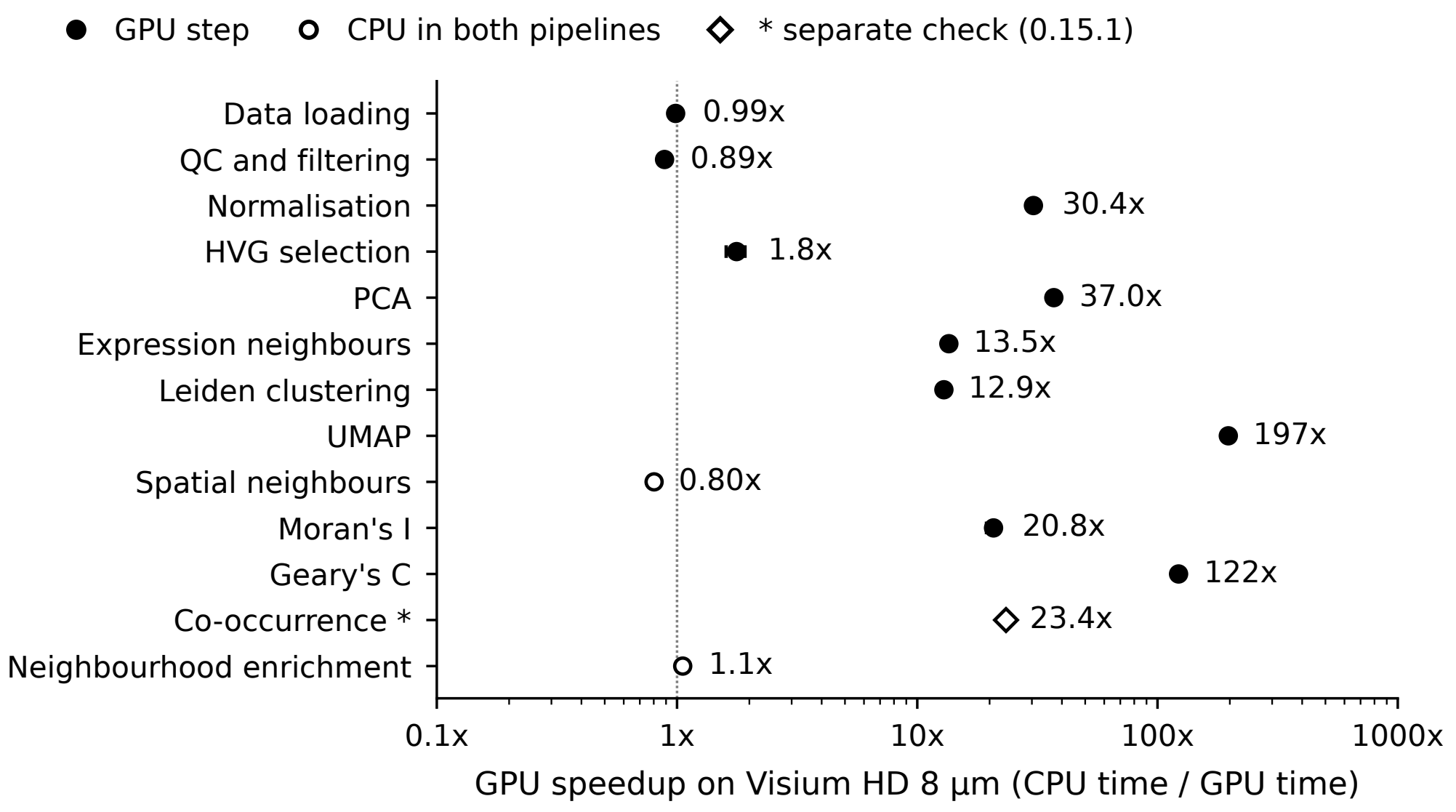
GPU step
CPU in both pipelines
* separate check (0.15.1)
Data loading
QC and filtering
Normalisation
HVG selection
PCA
Expression neighbours
Leiden clustering
UMAP
Spatial neighbours
Moran's I
Geary's C
Co-occurrence *
Neighbourhood enrichment
0.99x
0.89x
30.4x
1.8x
37.0x
13.5x
12.9x
197x
0.80x
20.8x
122x
23.4x
1.1x
0.1x
1x
10x
100x
1000x
GPU speedup on Visium HD 8 µm (CPU time / GPU time)